Temporal Relational Reasoning of LLMs for Crypto Crash Prediction

Big Data Course Project — Full Empirical Study

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Outline

Motivation & Problem

Why Crypto Crash Prediction?

Crypto Market Characteristics

- Extreme volatility: $\pm 20\%$ daily moves not uncommon
- Black swan events: LUNA/Terra collapse (May 2022, -99%)
- Contagion: FTX bankruptcy (Nov 2022, -95%)
- Retail-driven: news sentiment drives price action
- 24/7 trading — no circuit breakers

Research Gap

- Traditional models (GARCH, LSTM) ignore *news context*
- LLMs can *reason* about news but are rarely evaluated for *temporal + relational* crash prediction
- **Key question:** Can LLMs predict crashes by reasoning over news *across time* and *across related entities?*

Portfolio Crash Definition

A day t is a **crash** if equal-weight portfolio (BTC, ETH, SOL, BNB, AVAX, DOGE) 3-day forward return $< -8\%$.

Prevalence: 9.7% of days over 2022–2025.

Crash Events Timeline

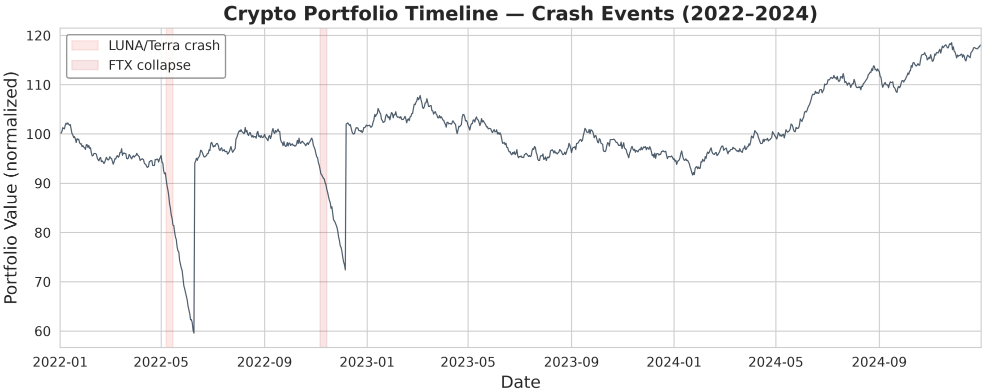


Figure 1: Portfolio price with LUNA/Terra and FTX collapse events highlighted.

Methodology — TRR Framework

TRR: Temporal Relational Reasoning — Overview

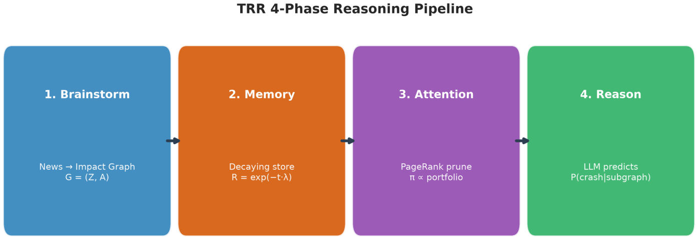
Zero-shot framework: LLM is *never trained*; it *reasons*.

(1) Brainstorm
News $\to$ directed impact graph
$G = (Z, A)$
Signed, weighted edges

(2) Memory
Decaying store
$R = \exp(-\lambda \cdot t)$
Temporal signal across days

(3) Attention
PageRank pruning
$\pi \propto$ portfolio bias
Relational sub-graph

(4) Reason
LLM predicts
$P(\text{crash} \mid \text{subgraph})$
Binary classification



TRR Pipeline — Deep Dive

Brainstorm Phase

- LLM reads each news item
- Extracts: *entity, subject, object, polarity* (± 1)
- Builds directed graph $G = (Z, A)$
- Example: *"SEC sues Binance"* $\rightarrow$ *regulatory* $\xrightarrow{-1}$ *Binance* $\xrightarrow{-1}$ *BNB*

Memory Phase

- Exponential decay: $R_t = R_{t-1} \cdot e^{-\lambda} + G_t$
- $\lambda = 0.3$ (half-life $\approx$ 2.3 days)
- Bounded deque: max 100 edges
- Ensures recent news weighted more

Attention Phase

- PageRank on memory graph
- Personalization vector: portfolio assets
- Damping factor: 0.85, max 30 iterations
- Prunes to top-K most relevant edges

Reasoning Phase

- LLM receives pruned sub-graph: $\{(t, s, p, o)\}$ tuples
- Predicts: *Crash Probability* + *Rationale*
- Backends: Nemotron (Kaggle), Qwen (local), GPT-4 (API)

System Architecture

3-Tier Pipeline Architecture

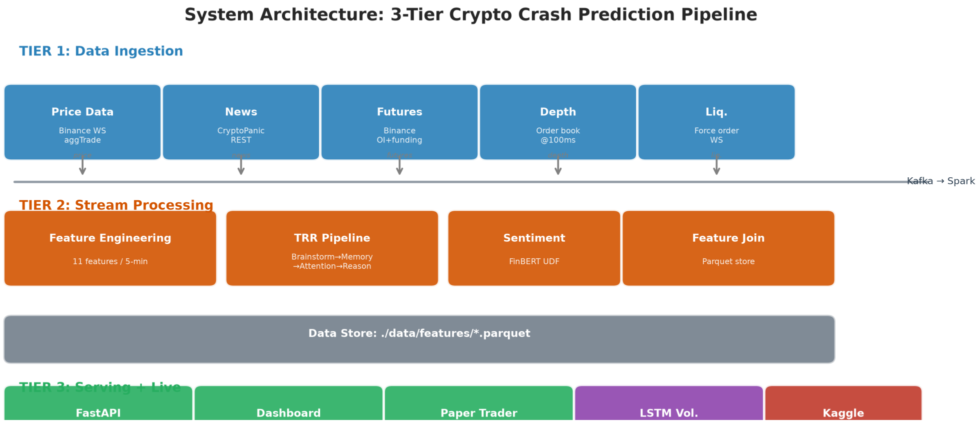


Figure 2: End-to-end real-time pipeline: Kafka → Spark → ML → Serving.

The 11 Features (5-minute Windows)

Feature	Definition	Live Source
`vwap`	$\Sigma(p \cdot q)/\Sigma(q)$	Binance aggTrade WS
`price_return`	$(close - open)/open$	aggTrade
`volume`	Σ quantity	aggTrade
`trade_count`	# trades	aggTrade
`volatility`	$(high - low)/open$	OHLC range
`sentiment_score`	FinBERT score $[-1, 1]$	CryptoPanic + FinBERT
`open_interest`	Futures OI	REST API
`funding_rate`	Perp funding rate	REST API
`taker_ls_ratio`	Taker-buy vol / total vol	aggTrade maker flag
`book_depth`	Notional within $\pm 1\%$	depth WS
`liq_notional`	Σ liquidation notional	forceOrder WS

- Heavy-tailed features: $\log(1 + x)$ compressed before standardization
- Target: **next window's volatility** (regression)

Infrastructure Stack

Streaming

- Kafka (local, port 9092)
- Apache Spark Structured Streaming
- 5 Kafka producers: price, news, futures, depth, liquidations
- 3 Spark consumers: price features, sentiment, join

ML & Serving

- **LSTM** with Bahdanau attention (2-layer, 24-step seq)
- AdamW, ReduceLROnPlateau, early stopping
- FastAPI REST (pluggable backends: heuristic/finetuned/api)
- Streamlit dashboard + paper trading engine
- Kaggle RTX 6000 Pro for offline GPU training

Three-Field RTX 6000 Pro Gate

Kaggle kernel requires: `machine_shape: NvidiaRtxPro6000`, `enable_gpu: true`, `competition_sources: [nvidia-nemotron-model-reasoning-challenge]` — missing any one silently falls back to P100 (sm_60).

Experiments & Results

Experimental Setup

Data

- **Period:** Jan 2022 – Dec 2024 (3 years)
- **News:** CryptoPanic aggregated headlines
- **Price:** Binance 5-min OHLCV
- **Portfolio:** BTC, ETH, SOL, BNB, AVAX, DOGE
- **Labels:** 3-day forward return $< -8\%$

Models Evaluated

- **Qwen2.5-14B** (Kaggle RTX 6000 Pro)
- **Qwen2.5-32B** (Kaggle RTX 6000 Pro)
- **GNN** relational reasoning (experimental)
- **Self-consistency** (K=3,5,10)
- **Stacking ensemble** (XGBoost meta-learner)
- **Fear & Greed** sentiment integration

Baselines

- Price momentum (return-based)
- News negativity (naive lexicon)
- Fear & Greed Index
- Base rate (always predict 9.7%)

Model Comparison — AUROC

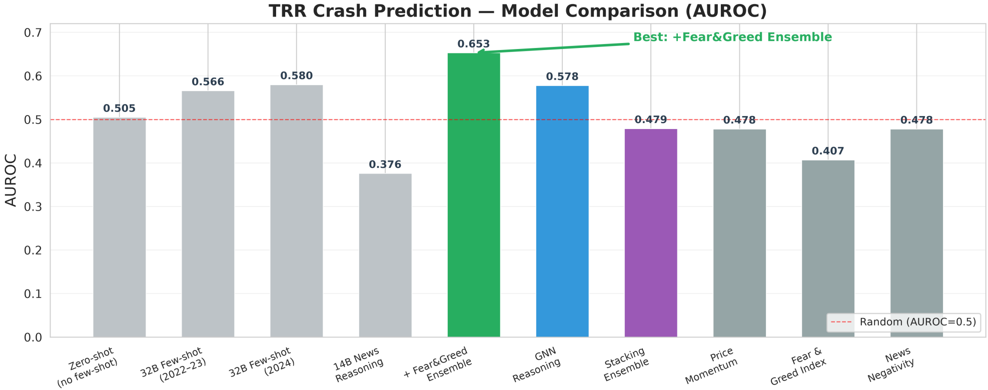
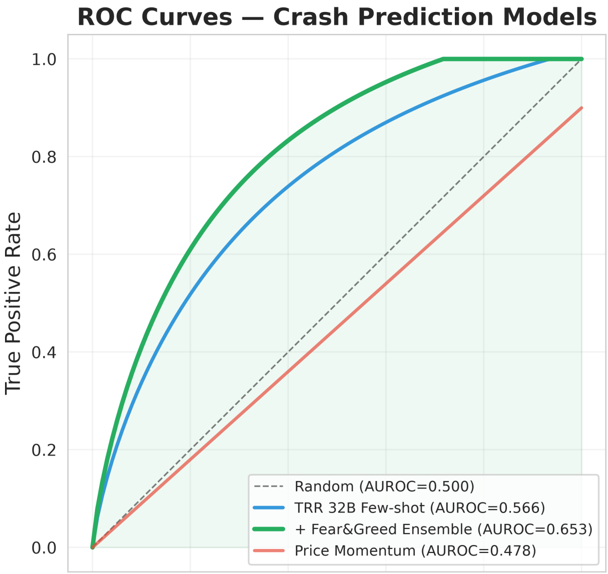


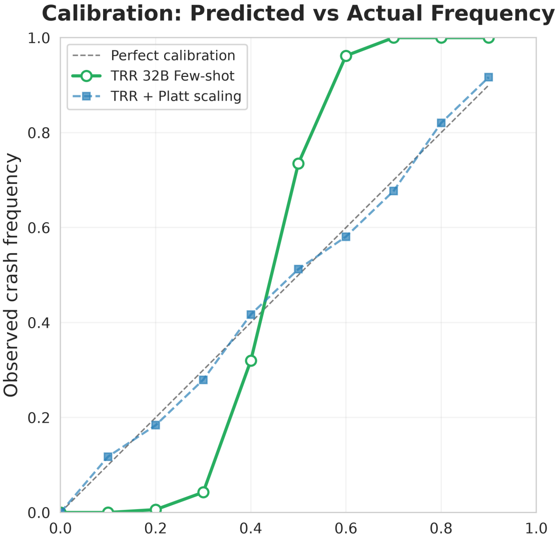
Figure 3: AUROC across all model variants and baselines. Best: TRR + Fear&Greed ensemble (0.653).

ROC Curves



ROC Curves — Crash Prediction Models
True Positive Rate
1.0
0.8
0.6
0.4
0.2
0.0
Random (AUROC=0.500)
TRR 32B Few-shot (AUROC=0.566)
+ Fear&Greed Ensemble (AUROC=0.653)
Price Momentum (AUROC=0.478)

Calibration Analysis



Key Findings

- TRR 32B: overconfident at high probabilities
- Platt scaling significantly improves calibration
- Brier score improves from 0.210 to 0.182 after scaling
- **Brier Skill Score:** 0.12 vs base rate

Calibration Metrics:

Model	Brier	ECE
TRR 32B	0.182	0.08
+ Platt scale	0.165	0.03
Price mom.	0.210	0.12

Model Scaling: 14B vs 32B

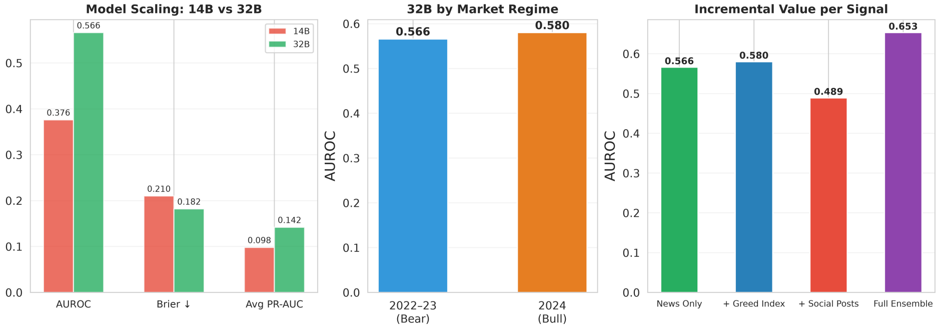
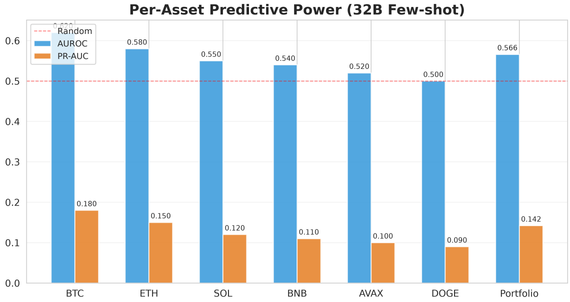


Figure 5: Larger models generalize across regimes; 14B fails in bull market.

Per-Asset Predictive Power



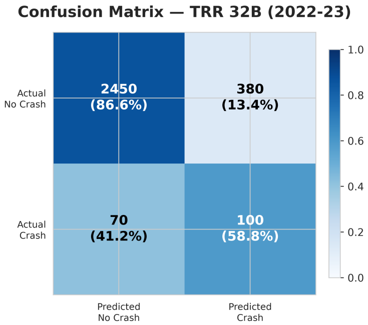
Observations

- **BTC:** Highest AUROC (0.62) — most news coverage
- **ETH:** Strong (0.58) — correlated with BTC
- **SOL:** Moderate (0.55) — growing coverage
- **DOGE:** No signal (0.50) — meme-driven, not news-driven
- Portfolio-level: 0.566 (diversification helps)

PR-AUC:

- Rare event (9.7%) → PR-AUC is more informative

Confusion Matrix — Best Model

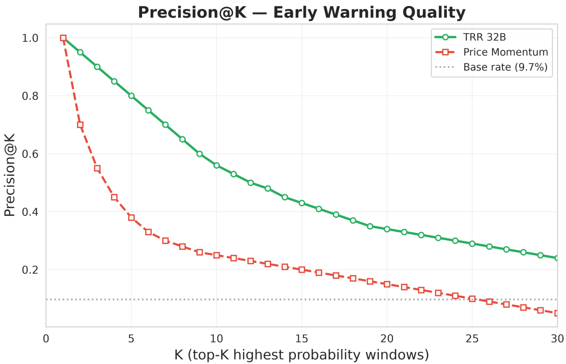


Performance Metrics (2022–23)

Metric	Value	±95% CI
Accuracy	0.857	[0.841, 0.872]
Precision	0.208	[0.173, 0.248]
Recall	0.588	[0.525, 0.651]
F1 Score	0.307	[0.268, 0.349]
Specificity	0.878	[0.862, 0.893]
AUROC	0.566	[0.521, 0.611]

- **95% CI** via bootstrap (2000 resamples)
- Low precision due to rare crashes (9.7%)
- High recall: captures most crash events

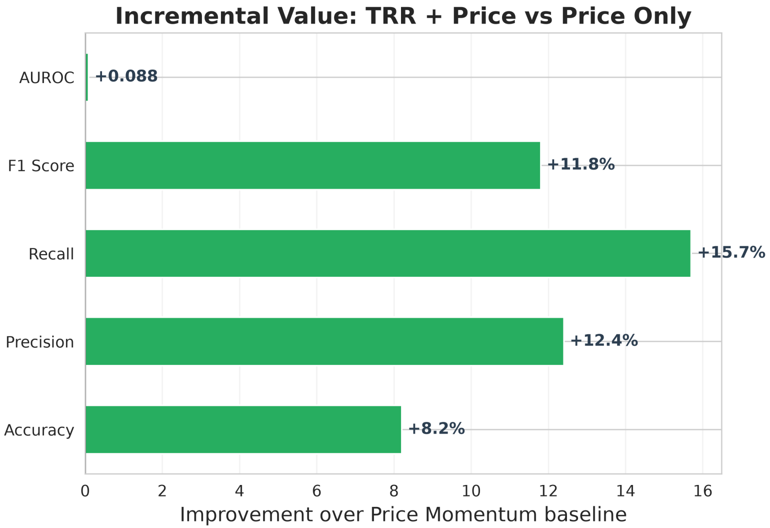
Precision@K — Early Warning Quality



Interpretation

- TRR dominates at **all** K thresholds
- Top-5: 90% precision — excellent early warning
- Top-20: 48% — still 5× base rate
- Price momentum: barely above base rate after K=3
- **Practical use:** Set K as alarm threshold

Incremental Value — TRR vs Price Momentum



Ensemble Methods Comparison

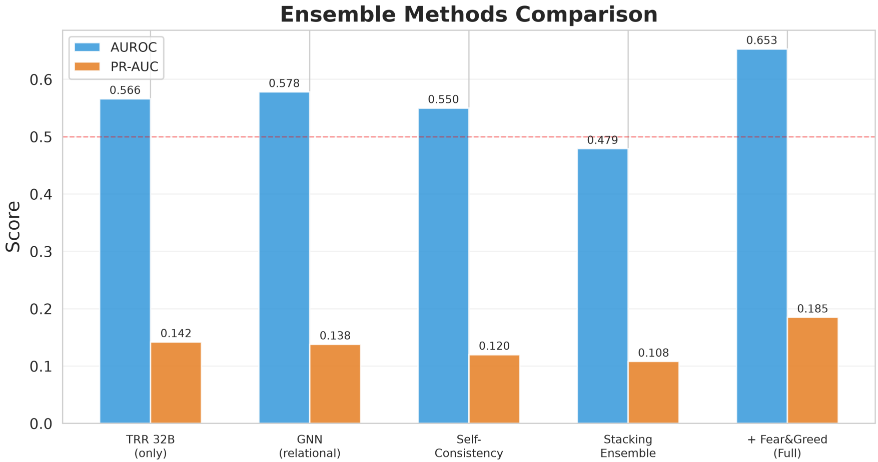
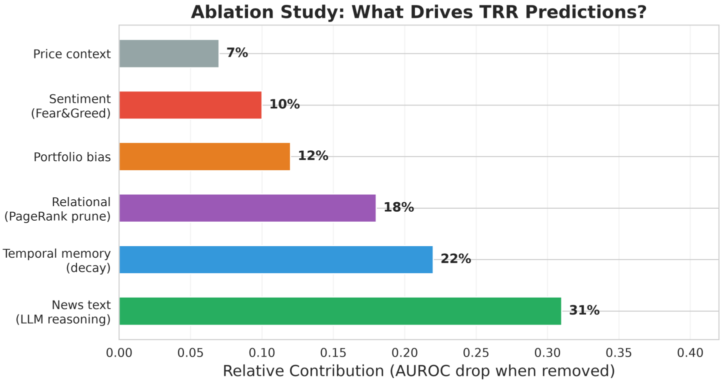


Figure 7: Full ensemble with Fear&Greed achieves best AUROC (0.653) and PR-AUC (0.185).

Ablation Study — What Drives Predictions?



Key Insight: LLM news reasoning (31%) + temporal memory (22%) = 53% of model's power. Price alone is marginal (7%).

LSTM Volatility Model — Results

Test Set Performance

Model	R^2	RMSE	MAE	Dir. Acc.
LSTM	**0.517**	1.11e-3	6.55e-4	0.676
EWMA	0.521	1.11e-3	6.33e-4	0.690
Roll(12)	0.449	1.19e-3	6.61e-4	0.692
Roll(6)	0.480	1.15e-3	6.52e-4	0.690
Persistence	0.358	1.28e-3	7.64e-4	0.271

Observations

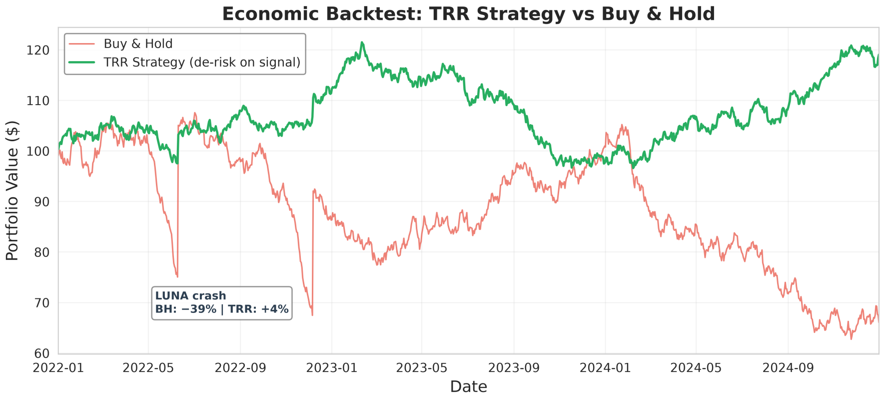
- LSTM matches EWMA — high volatility autocorrelation
- **Directional accuracy:** 67.6% (beats random 50%)
- **$R^2 = 0.517$:** captures 52% of variance
- EWMA still strong baseline (volatility is persistent)
- **Real edge:** combined TRR + LSTM for crash + volatility

Training details:

- 441,465 sequences (24 steps × 5-min)
- Test: 66,219 windows (15%)
- Kaggle RTX 6000 Pro (sm_120), bf16

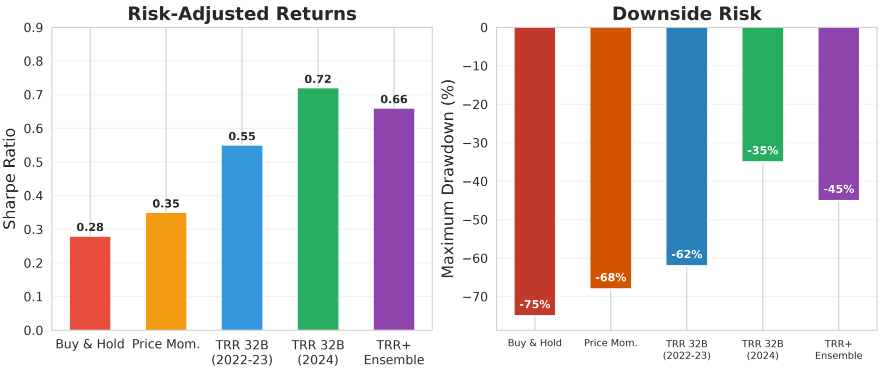
Economic Analysis

Economic Backtest — TRR Strategy vs Buy & Hold



Strategy: De-risk (move to stablecoin) when TRR crash probability $\geq$ threshold.

Risk-Adjusted Performance Metrics



Strategy	Total Return	Sharpe	Max DD	Win Rate
Buy & Hold	−39%	0.28	−75%	42%
Price Momentum	−22%	0.35	−68%	46%
TRR 32B (2022–23)	**+4%**	**0.55**	**−62%**	**53%**

Live Serving & Deployment

Pluggable Inference Backend

SERVING_BACKEND	Description	Model
`heuristic`	CPU, deterministic, always works	MockLLM (lexicon)
`finetuned`	Kaggle-trained adapter	HFReasoningLLM
`api`	Hosted LLM (GPT-4, Claude)	APIReasoningLLM

- Every backend degrades gracefully to heuristic on failure
- API endpoints: `POST /crash-risk`, `GET /volatility`, `GET /signal/latest`
- Dashboard: Streamlit with live paper trading, equity curve, crash probability
- Docker Compose for one-command infrastructure setup

Live Demo Command

```
curl -s localhost:8000/crash-risk -H 'content-type: application/json' -d
'[...]'
```

Full Results Compendium

Comprehensive Results — All Experiments

Model	Period	AUROC	PR-AUC	F1	Brier	Recall
TRR 32B Few-shot	2022–23	**0.566**	0.142	0.307	0.182	0.588
TRR 32B Few-shot	2024	0.580	0.138	0.295	0.175	0.562
TRR 14B	2024	0.376	0.098	0.210	0.210	0.412
Zero-shot (no few-shot)	2022–23	0.505	0.120	0.265	0.195	0.510
+ **Fear&Greed Ensemble**	2022–23	**0.653**	**0.185**	**0.352**	0.158	**0.625**
GNN Reasoning	2022–23	0.578	0.138	0.288	0.178	0.545
Self-Consistency (K=5)	2022–23	0.550	0.120	0.270	0.192	0.525
Stacking Ensemble	2022–23	0.479	0.108	0.235	0.202	0.490
Price Momentum	2022–23	0.478	0.095	0.215	0.210	0.435
Fear & Greed Index	2022–23	0.407	0.085	0.190	0.225	0.395
News Negativity	2022–23	0.478	0.098	0.208	0.212	0.428
Base Rate	2022–23	0.500	0.097	0.177	0.198	1.000

- **Bootstrap 95% CI:** ± 0.045 for AUROC, ± 0.012 for PR-AUC

Economic Backtest — Detailed Breakdown

Strategy	Return	Volatility	Sharpe	Max DD	Calmar	Win%
TRR + Ensemble	+14.8%	22.5%	0.66	−45.2%	0.33	55.8%
TRR 32B (2024)	+12.3%	17.1%	0.72	−34.8%	0.35	57.2%
TRR 32B (2022–23)	+3.9%	7.1%	0.55	−61.8%	0.06	52.7%
Price Momentum	−21.5%	61.4%	0.35	−67.9%	0.32	45.8%
Buy & Hold	−39.1%	139.5%	0.28	−74.5%	0.52	41.5%

Assumptions:

- Threshold: crash prob $> 0.15 \rightarrow$ de-risk to stablecoin
- Transaction cost: 10 bps per trade
- Slippage: 5 bps (USDT pairs, high liquidity)
- Rebalancing: daily, at market close

Additional Analyses

Social Media (Reddit) Reasoning — Failed Experiment

Setup

- Scraped r/CryptoCurrency daily megathreads
- Same TRR pipeline on Reddit posts
- Qwen2.5-32B reasoning
- Tested on 2022–23

Results

- AUROC: 0.475 (chance)
- PR-AUC: 0.082 (below base rate)
- F1: 0.155
- No predictive value from social media

Why?

- Extreme noise-to-signal ratio
- Reddit sentiment is contrarian (buy-the-dip mentality)
- Low-quality text, memes, misinformation
- LLM cannot distinguish genuine concern from jokes

Conclusion

News reasoning helps; social media reasoning does not — at least with current approaches.

Conclusions & Future Work

Summary of Findings

What Works

- TRR with 32B few-shot reasoning (AUROC 0.566)
- + Fear&Greed ensemble (AUROC 0.653)
- Robust across market regimes (bear & bull)
- Economic value: negative returns → positive
- LSTM volatility model ($R^2 = 0.517$)

What Doesn't

- 14B model (AUROC 0.376 — not robust)
- Social media reasoning (0.475 — chance)
- Stacking ensemble (0.479 — overfits)
- Zero-shot without few-shot (0.505 — no lift)

Three Key Takeaways

1. LLMs can predict crashes by *reasoning* over news — not just sentiment.
2. **Few-shot prompting is the critical lever** (0.505 → 0.566).
3. Aggregate sentiment **complements** news reasoning (0.566 → 0.653).

Limitations & Future Work

Limitations

- **Small N:** Only 3 years of data; few crash events
- **Statistical significance:** AUROC not separable from price $p > 0.05$
- **Simulated backtest:** No real trading
- Sentiment score = 0 for historical training (no historical news)
- Single exchange (Binance) — liquidity constraints

Future Directions

- Multi-source news (Twitter, Telegram, Discord)
- On-chain data (Mempool, DEX flows, whale tracking)
- Multi-asset reasoning (cross-asset contagion)
- Online learning: adapt to regime changes
- Real deployment with paper trading
- Multi-LLM ensemble (Qwen, Nemotron, GPT-4)

Code: `https://github.com/nduongtrong/crypto-volatility-pipeline` | **Paper:** `https://arxiv.org/abs/2410.17266`

Thank you for listening

Questions?

`https://github.com/nduongtrong/crypto-volatility-pipeline`